

Vatsal Maniar

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SUMMARY

MFE candidate with a B.S. in Computer Science; building derivatives pricing engines, MLE-calibrated stochastic risk models, and live options market-making systems. Core expertise in Monte Carlo simulation, VaR/CVaR, options Greeks, fixed income analytics, and systematic trading framework development.

EDUCATION

Stevens Institute of Technology

Hoboken, NJ

*M.S. Financial Engineering | GPA: 3.67/4.00 | Certificate: Financial Risk Engineering
Relevant Coursework: Market Microstructure, Computational Methods, Advanced Derivatives,
Stochastic Calculus, Pricing & Hedging, Risk Engineering & Management*

Sep 2025 – Dec 2026

Arizona State University

Tempe, AZ

B.S. Computer Science | GPA: 3.74/4.00 | Magna Cum Laude

Aug 2021 – May 2025

PROJECTS

ApexGP Markets – Quantitative Event Market Research Platform | Python, FastF1, Recharts

June 2026

- Built a quantitative event-market platform transforming 24 Formula One races into 500+ tradable binary contracts with dynamic probability repricing, race settlement, and season-long championship futures markets.
- Developed portfolio risk infrastructure with mark-to-market valuation, exposure limits (30% portfolio, 15% single-market), Sharpe analytics, drawdown monitoring, and full transaction audit across 24 events.

Hornet Contest – Live Derivatives Trading Competition | Excel VBA, Bloomberg

Jan 2026 – May 2026

- Engineered a VBA options market-making engine: automated real-time bid-ask quoting, Greeks-aware delta/gamma hedging, position booking, and intraday P&L attribution across multi-leg derivatives strategies in live Bloomberg trading sessions.
- Benchmarked Black-Scholes theoretical prices against live Bloomberg implied volatility quotes and bid-ask spreads; dynamically managed net delta, gamma, and vega exposure within defined risk limits throughout each active trading session.

Rates, FX & Options Pricing Engine | Python, Excel, Bloomberg, FRED

Oct 2025 – Dec 2025

- Calibrated GBM FX model on 30+ years of historical data; executed 20,000+ Monte Carlo simulations to estimate 99% VaR (~\$264K unhedged exposure) and priced European options benchmarked against Black-Scholes analytical solutions.
- Priced 24 Treasury bonds across 4 maturity buckets; bootstrapped yield curves, computed DV01/duration/convexity sensitivities, and validated delta-hedging ratios against Bloomberg and FRED market data.

EXPERIENCE

Alcove Ridge Consulting

Sep 2024 – Jun 2025

Software & Platform Development Intern

Remote

- Developed automated P&L reporting pipelines and investor dashboards tracking capital deployment and deal-level risk metrics across 10+ private market investments, reducing reporting latency by 35%.
- Built multi-asset scenario analysis and risk-adjusted return attribution models across 3 asset classes; automated cross module data reconciliation, compressing analyst review cycles by 45%.

ASU Capital Assets Management Office

May 2024 – May 2025

Asset Management Specialist

Tempe, AZ

- Reconciled financial records in \$1.5B institutional asset holdings, maintaining 99.9% data integrity and achieving 100% compliance across internal controls and sponsor audit requirements.
- Executed risk-sensitive inventory reviews across 500+ high-value assets (\$5K+), designing traceability documentation standards for \$5M+ in infrastructure and software assets.

ASU Admission Services

May 2022 – May 2024

Verification Supervisor

Tempe, AZ

- Led a 60-person verification team processing 30,000+ applicant records annually, designing productivity benchmarks that achieved department-record throughput at 98% data accuracy.
- Strengthened quality control and audit documentation procedures, ensuring regulatory alignment and zero compliance violations across 2+ years of compliance-sensitive workflows.

CERTIFICATIONS

GARP Financial Risk Manager (FRM) Part I Candidate | (Aug 2026)

TECHNICAL SKILLS

Languages: Python (NumPy, Pandas, SciPy, Matplotlib, QuantLib), C++, Java, R, MATLAB, VBA

Quantitative: Derivatives Pricing, Stochastic Calculus, Monte Carlo Simulation, Options Greeks, Backtesting, VaR/CVaR, Volatility Modeling, Factor Models (Fama-French), Fixed Income Analytics, Portfolio Optimization

Tools: Excel (Advanced Modeling), Bloomberg Terminal, Capital IQ, LSEG Workspace, GitHub, PowerBI